

Perspective on 'Delayed Yield'

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The phenomenon of 'delayed yield' has received wide attention in the literature during recent years. Considerable work has been devoted to the development of physically based mathematical models capable of successfully reproducing observed aquifer behavior. As a result of this effort, there appears to be at present a consensus of opinion among the main contributors to the theory of delayed yield as to an acceptable physical and mathematical interpretation of this phenomenon. In spite of this apparent consensus of opinion, there is still a considerable amount of controversy in the literature about some fundamental aspects of unconfined well hydraulics. The outstanding issues concern the physical assumptions underlying various delayed yield models, changes in the flow regime of an unconfined aquifer as a function of radial distance from the pumping well, the relationship between Boulton's α and other physical characteristics of the aquifer, the validity of conditions required in order to represent unconfined flow by integrodifferential equations, and the behavior of unconfined aquifers at large values of time. The present paper offers a perspective from the author's viewpoint on these and other related issues.

INTRODUCTION

In analyzing pumping test data from unconfined aquifers one often finds that the drawdown response fails to follow the classical Theis solution [Theis, 1935]. When the drawdowns are plotted versus time on logarithmic paper, they tend to delineate an inflected curve consisting of a steep segment at early times, a flat segment at intermediate times, and a somewhat steeper segment at later times. The early segment indicates that some water is released from aquifer storage instantaneously when drawdowns increase. The intermediate segment suggests the existence of an additional source of water which is released from storage with some delay in time. When most of the water is derived from this additional source, the time-drawdown curve becomes relatively steep again. This phenomenon has been traditionally referred to in the groundwater literature as 'delayed yield.'

The first mathematical model capable of mimicking delayed yield behavior was introduced into the literature by Boulton [1954b, 1963]. In his model, Boulton assumed that the amount of water released from storage per unit horizontal area of the aquifer owing to a unit average drawdown occurring at time τ is the sum of two components: S , a volume of water instantaneously released at time τ , and S_y , a volume of water the release of which is delayed in time according to the formula $\alpha S_y \exp[-\alpha(t - \tau)]$, where t is time ($t \geq \tau$) and α is a characteristic constant of the aquifer. Two other models which can be viewed as variations on the same theme were reported later by Boulton [1970] and Boulton and Pontin [1971]. The theory was used originally by Boulton [1963] and later by Prickett [1965] to describe a graphical procedure for determining the transmissivity T , storage coefficients S and S_y , and the so-called 'delay index' $1/\alpha$ of unconfined aquifers from pumping test data.

Although Boulton's model appears to fit field data quite well, it nevertheless fails to provide insight into the physical nature of the delayed yield phenomenon. Since the exponential rate of delay assumed by Boulton is hypothetical, his model is not related explicitly to the actual physics of the delay process. Furthermore, the model does not specify any definite relationship between α (controlling the rate of delay) and other physical characteristics of the aquifer system. This is not to imply that Boulton neglected to speculate about the physical nature

of delayed yield. On the contrary, he postulated three possible mechanisms for the process. In his early paper, Boulton [1954b] attributed delayed yield to leakage either from (1) a 'bed of fine-grained material' overlying the aquifer and containing the water table, or from (2) 'fine-grained compressible' layers interbedded with the aquifer material. In the first case [Boulton, 1954b, p. 473],

As pumping proceeds and the pore-water pressures at the base of the fine sand are lowered, water slowly drains out of this bed giving a delayed yield from storage.

In the second case [Boulton, 1954b, p. 472],

If an artesian aquifer contains beds of fine grained compressible material, these beds do not compress immediately when the pore-water pressures at their boundaries are reduced by pumping. Thus, water is squeezed out of them rather slowly, causing a delayed yield from storage similar to that produced by the slow draining of fine-grained beds under water-table conditions.

Only in his second paper on the subject has Boulton [1963] attributed delayed yield to water table conditions in the aquifer itself. However, his physical explanation for the phenomenon was vague [Boulton, 1963, p. 470]:

... when a well is pumped at a steady rate in a homogeneous aquifer under water-table conditions, the water-bearing material through which the water-table has fallen during the early stages of pumping does not immediately yield up its water. ... Moreover, during this early period of pumping, which may be only a few minutes, the aquifer compresses elastically and the water itself expands in response to the reduction in hydrostatic pressure. ... The small elastic response is similar to that which is associated with non-equilibrium conditions in confined or artesian aquifers. If there were no delay in yield from above the water table, the coefficient of storage representing the elastic compression of the aquifer and expansion of the water would be of no significance. ... Owing, however, to the delayed yield to the water-table, the very early time-drawdown curve follows the Theis curve for an artesian aquifer. However, as the yield to the water-table becomes effective, the time-drawdown curve becomes a 'leaky-artesian aquifer curve,' the drawdown attaining temporary equilibrium (or near equilibrium). After a longer time the specific yield attains an almost constant value and the time-drawdown curve then merges with the Theis non-equilibrium curve associated with the coefficient of storage for water-table conditions.

If the aquifer does not yield any water by gravity drainage during the early artesian stage, how can the water table fall during this stage? If the water table is actually falling while the thickness of the initial saturated zone remains unchanged, the

top of the saturated zone must be under negative pressure. This could happen if, for example, the entry of air from above would be precluded by a layer of high water content as suggested recently by *Bouwer and Rice* [1978]. Suppose that the top of the saturated zone is indeed under negative pressure. Would this provide a sufficient explanation for delayed yield? If so, would it still be necessary to account for water release from storage by an elastic artesian effect as suggested by *Boulton*?

The question whether negative pressure at the top of the saturated zone could provide an explanation for delayed yield can be answered only by analyzing the effect of flow in the unsaturated zone on aquifer behavior during the process of gravity drainage. *Boulton's* [1954b, 1963] original papers on delayed yield do not make any explicit mention of an 'unsaturated zone' above the water table. Furthermore, *Boulton's* mathematical model of exponential yield is hypothetical and as such it does not make any use of unsaturated flow theory. In spite of this, *Boulton's* vague reference to a water table which is falling faster than the ability of the aquifer material to yield water has caused many authors to believe that the key to the understanding of delayed yield lies in the unsaturated zone [Youngs and Smiles, 1963; Vachaud, 1968; Vachaud and Thony, 1969; dos Santos and Youngs, 1969; Youngs, 1969; Cooley, 1971]. More recently, *Bouwer and Rice* [1978] have suggested that delayed yield could be related to delayed air entry into the unsaturated zone immediately above the water table.

However, all the available evidence indicates that the gravity drainage phase of drawdown response in unconfined aquifers is not seriously affected by unsaturated flow, at least in the absence of air entry effects. Numerical studies by *Brutsaert* [1970] and *Brutsaert et al.* [1971] were the first to show that in aquifer flow the unsaturated 'cone of depression' is usually insignificant and '... The overall drawdown ... is too slow for capillarity to be a critical factor.' A few years later, *Streltsova* [1973], who was relying on field experiments by *Meyer* [1962], demonstrated the subordinate role that unsaturated flow plays in the delayed yield process. However, the most definitive study about the role of unsaturated flow in the gravity drainage phase of delayed yield has been published more recently by *Kroszynski and Dagan* [1975]. They have developed an analytical solution for flow to a well in a rigid aquifer under saturated-unsaturated conditions. Their analytical solution has been verified numerically with the aid of a finite element program slightly modified after *Neuman* [1975b] and *Neuman et al.* [1974]. Both the analytical and numerical results clearly indicate that unsaturated flow has very little influence on the time-drawdown response of unconfined aquifers during gravity drainage. The effect of air entry will be discussed later in the text.

The first answer to the dilemma of delayed yield based on clearly defined physical concepts has been offered simultaneously and independently by *Neuman* [1972, 1973] and *Streltsova* [1972a, b, 1973]. Both of these authors treated the unconfined aquifer as a compressible system and the water table as a moving material boundary, thus disregarding both air entry effects and unsaturated flow. They showed that delayed yield can be simulated mathematically by treating the specific storage and specific yield as constants. This differs from *Boulton* who allowed the specific yield to vary exponentially with time. Both authors demonstrated that the delayed yield phenomenon may be caused by a time lag between the early artesian response of the aquifer and the subsequent downward movement of the water table due to gravity drainage. This viewpoint

was reflected in the title of *Neuman's* paper which referred to 'delayed response of the water table' instead of the traditional 'delayed yield.' In his subsequent papers on the subject, *Neuman* [1974, 1975a] suggested replacing the term 'delayed yield' by 'delayed gravity response.'

Although *Streltsova* [1972a, b, 1973] and *Neuman* [1972, 1973] shared a similar outlook about the physics of the problem, their mathematics differed as a result of differences in the level of approximation. *Streltsova* worked only with two drawdowns at each point in the horizontal plane: (a) the average drawdown $\bar{s}$ along the vertical, defined by

$$\bar{s} = \frac{1}{b} \int_0^b s \, dz \quad (1)$$

and (b) the drawdown s_{wt} of the water table. In this way, the problem is reduced from three- to two-dimensional radial flow in the horizontal plane. When pumping starts, the sudden drop of the water level in the well causes a reduction in the average head of the aquifer due to an artesian effect. Since dewatering cannot occur instantaneously, the drawdown at the water table remains initially less than the average drawdown. This difference between $\bar{s}$ and s_{wt} results in vertical gravity drainage. *Streltsova* introduced a further simplification by assuming that the rate of vertical gravity drainage is linearly proportional to the difference between $\bar{s}$ and s_{wt} , i.e.,

$$\partial s_{wt} / \partial t = \alpha (\bar{s} - s_{wt}) \quad (2)$$

where α is a constant that represents vertical diffusivity and is given by

$$\alpha = 3K_z / S_y b \quad (3)$$

In this way the effect of gravity drainage becomes analogous to that of leakage. The analogy made it possible for *Streltsova* to derive approximate expressions for $\bar{s}$ and s_{wt} as functions of radial distance from the pumping well and time. Her expression for the average drawdown $\bar{s}$ is identical to that of *Boulton* [1954b, 1963]. The only difference lies in the physical meaning assigned by *Streltsova* to α in terms of K_z , S_y , and b as given by (3). The expressions for s_{wt} , on the other hand, differ substantially from those given for $\bar{s}$ by *Boulton* [*Streltsova and Rushton*, 1973].

The approach of *Neuman* [1972, 1973] differs from that of *Streltsova* in two respects: (a) it requires fewer simplifying assumptions, and (b) it allows the drawdowns to vary continuously with the vertical coordinate, thereby retaining the true three-dimensional axisymmetric character of the flow regime. *Neuman's* approach differs from an earlier work by *Boulton* [1954a], which did not consider delayed yield, in that it takes into account aquifer compressibility. Since the resulting problem is nonlinear, *Neuman* used a perturbation technique to obtain a first-order linearized solution for drawdown at each point in the aquifer as a function of time. The linearization is valid as long as the drawdown at the water table remains small in comparison with the initial saturated thickness of the aquifer. The time-drawdown curves for points lying below the water table exhibit the typical inflection pattern associated with delayed yield. The same holds true for the average drawdown $\bar{s}$ which is obtained simply by integrating the point-drawdown solution along the saturated thickness.

The three-dimensional nature of *Neuman's* approach enabled him to extend his theory to the case of partially penetrating wells [*Neuman*, 1974]. The results were used to develop a method for the analysis of pumping test data from uncon-

finned aquifers exhibiting delayed yield in the presence of fully or partially penetrating wells [Neuman, 1975a].

Neuman's approach of treating unconfined aquifers as compressible units while at the same time allowing the drawdown to vary along the vertical has later been adopted by Streltsova [1974] in lieu of her original theory. The same approach has also been adopted by Boulton and Streltsova [1975] and generalized by them to the case where the water table is located in an aquitard overlying a pumped aquifer. In the particular case where there is no aquitard, their solution reduces to that of Neuman [1974]. This seems to indicate that Boulton, Streltsova, and Neuman have come to a consensus of opinion as to a possible physically based mathematical model of delayed yield. A similar viewpoint dominates the more recent work of Herrera *et al.* [1978] in which the problem is cast in integrodifferential form.

In spite of this apparent consensus of opinion among the main contributors to the theory of delayed yield, certain fundamental aspects of this theory are still the subject of debate in the literature. For example, there appears to be some confusion in the literature about the physical assumptions underlying various delayed yield models now in existence. On the other hand, the more recent work of Bouwer and Rice [1978] raises a question about the nature of the early storage coefficient in these physical models and suggests an additional mechanism for the release of water from storage at early time. Another study by Gambolati [1976] leads to some questions which concern changes in the flow regime of an unconfined aquifer as a function of radial distance from the pumping well, and it shows that the behavior of unconfined aquifers at large values of time requires further investigation. This latter study also focuses on, and adds to, an ongoing controversy about the relationship between Boulton's α and other physical characteristics of the aquifer [Streltsova, 1976; Neuman, 1976; Gambolati, 1976]. In particular, Gambolati tries to assign a 'true' physical meaning to α , and there is a serious question about our ability to define a 'true' α in light of new theoretical results obtained recently by Herrera *et al.* [1978]. Their results include the derivation of mathematical conditions that must be satisfied if one wants to represent unconfined flow by integrodifferential equations, and there is a need to investigate the extent to which such conditions are met in nature. The remainder of this paper is devoted to a discussion of these issues.

RELIABILITY OF TIME-DRAWDOWN ANALYSES USING DIFFERENT DELAYED YIELD MODELS

In a recent publication, Ehlig and Halepaska [1976] have investigated numerically the case of a confined aquifer which becomes unconfined in the vicinity of the pumping well. An analytical solution to this confined-unconfined problem has been originally developed by Moench and Prickett [1972]. In their numerical study, Ehlig and Halepaska were able to verify the analytical solution of Moench and Prickett. Both the analytical and the numerical results indicated that, under confined-unconfined conditions, a log-log curve of average drawdown versus time may exhibit an inflection point similar to that associated with delayed yield in strictly unconfined aquifers. The numerical study showed that inflections in the time-drawdown curve may also occur because of sharp discontinuities in the material properties of the confined-unconfined aquifer, or because of the effect of impermeable lateral boundaries. When linear leakage from above was included in the model, its effect was merely to reduce the slope of the late time-

drawdown curve without causing an inflection in this curve.

Ehlig and Halepaska [1976] have also investigated the effect of delayed yield in the unconfined portion of a confined-unconfined aquifer on its time-drawdown response. The effect of delayed yield was simulated by allowing the specific yield in their model to vary exponentially with time in a manner analogous to that of Boulton [1954b, 1963] in his early work. As expected, this produced an inflection in the time-drawdown curve even when the aquifer was strictly unconfined.

From these results, Ehlig and Halepaska [1976] concluded that pumping test data exhibiting an inflection in the time-drawdown relationship cannot be interpreted uniquely without prior knowledge about the nature of the aquifer. This conclusion is obviously correct in the sense that a method of interpretation developed specifically for unconfined aquifers should not be applied to a confined-unconfined system, and vice versa. Unfortunately, Ehlig and Halepaska viewed their conclusion in a much broader sense, implying that a reliable estimate of aquifer characteristics cannot be obtained even if the aquifer is known to be homogeneous and strictly unconfined. This was no longer based on the results of their study, but rather on the erroneous belief that of all the analytical solutions for delayed yield in unconfined aquifers [Ehlig and Halepaska, 1976, p. 1175]

... each ... requires a different interpretation. Boulton's [1954b] delayed yield theory uses an exponentially decreasing source term to account for water released from storage in the unsaturated zone. Neuman [1972] assumes that the contribution from the unsaturated zone is negligible and achieves the desired curves by assuming that the free surface is a moving material boundary. Streltsova [1973] suggests that the inflection is due to a discontinuity in the potential at the well face when pumping commences which is gradually reduced as the system reestablishes equilibrium.

The authors therefore concluded that [Ehlig and Halepaska, 1976, p. 1182]

... if the aquifer is known to be strictly unconfined and consists of a material, such as fine sand, which tends to retain water above the saturated zone, then Boulton's theory, which assumes significant contribution from the unsaturated region, may be most appropriate. If instead the strictly unconfined aquifer is a coarse sand, Neuman's theory makes the correct assumptions. If a discontinuity in the potential at the well face causes significant delay in the response, then Streltsova's theory may be the best choice. ... Boulton's and Neuman's interpretations are mutually exclusive on a theoretical basis.

By now the reader should be aware from my introduction that (a) Boulton's [1954b, 1963] exponential source term is 'operational' and therefore free of any specific assumptions about the physics of delayed yield; (b) unsaturated flow plays an insignificant role in the mechanics of delayed yield, even in the case of fine sands [Kroszynski and Dagan, 1975]; (c) the models of Neuman [1972, 1973, 1974] and Streltsova [1972a, b, 1973] are based on an identical interpretation of the physical processes that cause delayed yield; (d) therefore both of these models imply a discontinuity between the average potential along the well face and the water table potential, as can be easily verified from an examination of Neuman's results; and (e) the models of Neuman and Streltsova differ from each other primarily in their level of mathematical sophistication and accuracy, one maintaining the three-dimensional character of the problem, the other reducing it to two dimensions by relying on simplifying assumptions.

Furthermore, Boulton's model and the models of Neuman and Streltsova are by no means mutually exclusive on physical

grounds, although, as will be shown later, they are not necessarily consistent from a mathematical point of view. In fact, Boulton's model can be easily reconciled with those of Neuman and Streltsova simply by providing appropriate expressions for α in terms of other physical parameters as well as the space and time coordinates. For example, *Streltsova* [1972a, b, 1973] demonstrated very clearly that her model for the average drawdown becomes identical to that of Boulton when α is set equal to $3K_z/S_y b$ as in (3). This means that Boulton's model can be derived exactly on the basis of Streltsova's physical assumptions, which completely disregard flow in the unsaturated zone. Similarly, *Neuman* [1975a] was able to show that his solution for the average drawdown can be closely reconciled with that of Boulton merely by setting

$$\alpha = (K_z/S_y b)[3.063 - 0.567 \log(\beta)] \quad (4)$$

within the range $0.01 \lesssim \beta \lesssim 5.00$. Later, *Gambolati* [1976] showed that the two solutions can be reconciled exactly for all values of β by redefining α in terms of the same physical parameters as manifested in his Figure 6. Since the unsaturated zone is disregarded in Neuman's model, it is clear that contrary to *Ehlig and Halepaska* [1976], Boulton's 'exponentially decreasing source term' does not necessarily 'account for water released from storage in the unsaturated zone.' Instead, it seems to account primarily for water released from storage in the saturated zone immediately below the water table as the latter drops.

Concerning the uniqueness of determining aquifer characteristics from time-drawdown data, *Neuman* [1975a] has shown that for fully penetrating wells, his method and that of *Boulton* [1963] and *Prickett* [1965] yield identical values of T , S , and S_y (or K_r , S_s , and S_y , if b is known). Since *Streltsova's* model for the average drawdown is identical to that of Boulton (except that her α is given by $3K_z/S_y b$), it will also yield the same numerical values for these parameters. On the other hand, *Streltsova's* definition of α differs from that of Neuman, and therefore the two models will yield different values for K_z (whereas Boulton's model will only yield a value for the lumped parameter α). The difference between the values of K_z stems merely from the fact that Neuman's model is mathematically more accurate than *Streltsova's* and not from any fundamental differences in the underlying physical assumptions.

When the pumping well is partially penetrating, the earlier models of *Boulton* [1954b, 1963] and *Streltsova* [1972a, b, 1973] no longer apply, and the data must be analyzed by the method of *Neuman* [1975a]. This again stems merely from the three-dimensional character of Neuman's model and not from any differences in the underlying physics.

ROLE OF DELAYED AIR ENTRY AT EARLY TIME

All of the existing delay yield models assume that at early time, water is released from storage primarily by elastic compression of the aquifer material and expansion of the water. As a result of this artesian effect, drawdowns everywhere below the water table fall on the Theis curve at sufficiently small values of time.

Recently, the viewpoint that artesian effects are the primary mechanism of water release at early time has been questioned by *Bouwer and Rice* [1978]. Their reason for questioning this view is related to the fact that many pumping tests yield values of $\sigma = S/S_y$ that range from 0.001 to 0.01 or more. Such relatively large values of σ imply that in unconfined aquifers the early storage coefficient S is larger than the artesian storage coefficient of most confined aquifers. *Bouwer and Rice* suggest

that this apparent contradiction could be resolved by viewing S not as a purely artesian storage coefficient but rather as a storage coefficient that arises from the combined action of artesian and air entry effects. In their view, if the movement of air from the atmosphere toward the water table is restricted (say, by a saturated layer of relatively low permeability within the unsaturated zone), the water table may initially start falling while the saturated thickness of the aquifer remains essentially unchanged. This implies that the top of the saturated zone must be under negative pressure, the absolute value of which increases as the water table (defined as the zero pressure isobar) continues to drop. This process lasts until air pressure in the unsaturated zone becomes sufficiently negative to create a situation whereby air can flow freely from the atmosphere toward the water table (say, by causing enough water to drain from the overlying saturated layer until the latter no longer acts as a barrier to the flow of air). Once direct contact between the atmosphere and the top of the saturated zone has been created in this manner, gravity drainage can then proceed unhindered at the full rate compatible with the specific yield of the aquifer.

There is little doubt that air entry effects of the kind described by *Bouwer and Rice* [1978] may indeed occur in nature and perhaps play a role in the delayed yield process. However, the importance of this role in comparison to that of artesian storage is not entirely clear at the present time and will require further investigation. Prior to discussing the air entry concept further, it is important to recognize that there are at least two reasons why one should expect to find higher artesian storage coefficients under unconfined conditions than under confined conditions. The first reason is due to the presence of entrapped air bubbles or dissolved air in the water when the aquifer is unconfined. Although the compressibility of water is generally very low, the magnitude of this compressibility can increase significantly when the water includes isolated bubbles of air and air in solution. That this compressibility alone is sufficient to generate a distinct delayed yield effect has been demonstrated very clearly in the numerical experiments performed by *Brutsaert et al.* [1971, Figure 7]. The second reason why the elastic storativity of unconfined aquifers could be relatively high stems from the fact that many of these aquifers consist of shallow unconsolidated materials with a significant clay content. In general, one should expect the compressibility of such alluvial deposits to increase as the degree of consolidation decreases and the content of clay increases.

Since both the water and the porous matrix tend to be more compressible under unconfined than under confined conditions, the finding that S is relatively large in unconfined aquifers appears to be consistent with existing delayed yield theory which views S as an elastic storage coefficient. Although S could perhaps also be related to air entry effects, as suggested by *Bouwer and Rice* [1978], this has not yet been demonstrated by a theoretical model that would be capable of mimicking delayed yield behavior without considering the effect of compressibility. Unfortunately, the experimental model constructed by *Bouwer and Rice* [1978, Figure 8] was also unable to clearly reproduce the typical inflection in the time-drawdown curve which is so familiar from field experiments. This means that there is neither theoretical nor experimental evidence to support the view that air entry effects could actually cause an inflection in the time-drawdown curve, as delayed yield would demand. Furthermore, the air entry concept leads to several conceptual difficulties which are at least as severe as those associated with the existing delayed yield theory. One of

these difficulties is related to the fact that delayed yield phenomena do not always appear to be restricted to situations where the aquifer is overlain by a saturated fine-grained layer; in the absence of such a layer there is no air entry effect, and therefore S must have a different physical meaning than that assigned to it by Bouwer and Rice.

Another conceptual difficulty is related to the hydraulic properties of fine-grained layers that may overlie an unconfined aquifer: according to Bouwer and Rice [1978], once negative pressure above the water table in the aquifer has reached a critical value, water from the overlying fine-grained sediments can drain into the aquifer, thus allowing air to flow through these sediments so as to raise the air pressure above the water table to atmospheric. However, such fine-grained sediments will usually have a relatively low permeability to water, and if they are also relatively thick, considerable time may be required until enough water will drain out of them to enable free passage of air into the aquifer. Even after such a passage has been created, the relative permeability to air will remain quite low until considerable amounts of water have drained out of the fine-grained sediments. Thus the establishment of atmospheric air pressure in the aquifer may require much more time than the actual time (usually minutes) at which the flat portions of the time-drawdown curves develop in typical field situations. It should be mentioned here that in the soil column experiments reported by Bouwer and Rice, the fine-grained sediments were represented by a layer of glass beads having a permeability only 10 times less than the underlying 'aquifer' material and a thickness of only 4 cm.

A third conceptual difficulty associated with the air entry concept has to do with the fact that according to available field data [Neuman, 1975a], the delay yield process appears to be fully reversible. In other words, when pumping stops and drawdowns are allowed to recover, the rate of recovery corresponds to a fully reversible process. This is compatible with the notion of elastic storage, but is generally not compatible with the air entry concept of Bouwer and Rice, owing to a strong hysteresis effect in the water retention characteristics of unsaturated sediments.

Since air entry effects have not yet been shown to cause an inflection in the time-drawdown behavior of unconfined aquifers, and since the concept itself is not entirely clear, I am led to conclude that this concept does not offer any theoretical advantage over the artesian storage theory at the present time.

INFLUENCE OF RADIAL DISTANCE ON FLOW REGIME

In a recent analysis of theoretical works related to delayed yield, Gambolati [1976] has concerned himself with the question to what extent is the nature of the flow regime influenced by radial distance from the pumping well. Based on an examination of time-drawdown behavior at 'a fixed distance from the well,' Gambolati assumed that the time at which radial or vertical flow start to predominate [Gambolati, 1976, p. 36]

... [is not] the same ... everywhere within the aquifer, and hence we cannot characterize the overall aquifer behavior by a unique definition. At any instant, three cylindrical regions may be recognized within a water table aquifer. In the inner region, flow is radial, the Theis solution holds with a total storage coefficient equal to $S_s + S_y \approx S_y$, and the contribution to the rate of withdrawal is small. In the middle region, flow is both horizontal and vertical, the Theis solution does not apply, and the contribution to the discharge is predominant ... [for it] comes mainly from the free surface. In the outer region, flow is again radial, the drawdown conforms with the Theis solution with

respect to t_s , and the contribution to flow is not very important for it comes from an elastic storage which is much smaller than the specific yield [provided S_y/S is large]. The idealized vertical surfaces separating these three regions move in time toward the lateral boundary, but at the beginning, immediately after the start-up they coincide with the line sink, and only the third type region exists ... at any time there is an aquifer zone reacting as if it were artesian ... [vertical flow] exerts its influence at any but the initial time somewhere within the aquifer (in the second type region).

This viewpoint is reminiscent of Prickett's [1965, p. 9] assumption that 'the cone of depression is distorted' during the time when delayed yield is important.

However, theory indicates that none of these assumptions are valid in the general case. From Figure 7 of Neuman [1972] we know that the cone of depression is not distorted, and its slope varies monotonically with radial distance. In other words, the cone of depression does not exhibit an inflection of the type observed in time-drawdown curves.

To investigate further the influence of radial distance on the flow regime, let us consider Neuman's solution for the average drawdown in the case of fully penetrating wells. According to Neuman [1972, 1973, 1975a], this solution is given by

$$\bar{s}(r, t) = \frac{Q}{4\pi T} \int_0^\infty 4x J_0(x\beta^{1/2}) \left[w_0(x) + \sum_{n=1}^\infty w_n(x) \right] dx \quad (5)$$

where

$$w_0(x) = \frac{\{1 - \exp[-t_s\beta(x^2 - \gamma_0^2)]\} \tanh(\gamma_0)}{[x^2 + (1 + \sigma)\gamma_0^2 - [(x^2 - \gamma_0^2)^2/\sigma]]\gamma_0} \quad (6)$$

$$w_n(x) = \frac{\{1 - \exp[-t_s\beta(x^2 + \gamma_n^2)]\} \tan(\gamma_n)}{[x^2 - (1 + \sigma)\gamma_n^2 - [(x^2 + \gamma_n^2)^2/\sigma]]\gamma_n} \quad (7)$$

and the quantities γ_0 and γ_n are the roots of the equations

$$\sigma\gamma_0 \sinh(\gamma_0) - (x^2 - \gamma_0^2) \cosh(\gamma_0) = 0 \quad (8)$$

$$\gamma_0^2 < x^2$$

$$\sigma\gamma_n \sin(\gamma_n) + (x^2 + \gamma_n^2) \cos(\gamma_n) = 0 \quad (9)$$

$$(2n - 1)\pi/2 < \gamma_n \leq n\pi \quad n \geq 1$$

From (5)–(7) it is seen that the relationship between the average dimensionless drawdown $\bar{s}_D$, and dimensionless time, t_s or t_y , is a function of the dimensionless parameters β and σ . Figure 1 illustrates these dimensionless curves for the particular case where $\sigma = 10^{-4}$ and a range of β values between 0.001 and 7.0. In order to obtain the effect of radial distance, we note that $\beta t_s = K_s t / Sb$, i.e., if βt_s is kept constant for a given aquifer, t is also constant. One can therefore develop dimensionless distance-drawdown curves merely by tracing the loci of points corresponding to constant values of βt_s in Figure 1. Four such distance-drawdown curves are shown for $\beta t_s = 10^0, 10^1, 10^2$, and 10^3 . In addition, the early Theis curve corresponds to the limiting case where βt_s tends to zero, whereas the late Theis curve represents the limiting distance-drawdown curve for $\beta t_s \rightarrow \infty$.

A striking feature of the distance-drawdown curves in Figure 1 is that, contrary to the time-drawdown curves, they do not exhibit a point of inflection. Originally, this observation was made some time ago by Neuman [1975a] in connection with the analysis of distance-drawdown data from unconfined aquifers. It means that just as in the case of the drawdown of the water table, the slope of the average drawdown also varies monotonically with radial distance from the pumping well.

The distance-drawdown curve for very early time (small βt_s)

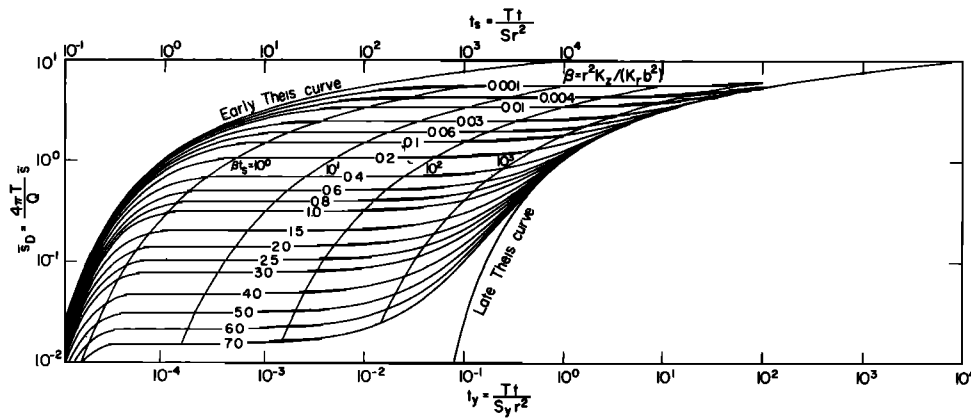


Fig. 1. Dimensionless time-drawdown (β -constant) and distance-drawdown ($\beta t_s = \text{constant}$) curves for $\tau = 10^{-4}$.

corresponds to the early Theis curve. This indicates that immediately after pumping starts the entire aquifer reacts as if it were artesian, and vertical hydraulic gradients are insignificant.

The distance-drawdown curve for late time (large βt_s) corresponds to the late Theis curve. This means that the entire aquifer behaves as if the Dupuit assumptions were correct, thus implying that flow is primarily horizontal and the effect of vertical gradients is again negligibly small. In other words, at small values of time the entire aquifer is subject to an artesian flow regime, whereas at large values of time the entire aquifer is subject to a regime conforming to the Dupuit assumptions. In neither case can one talk about the existence of distinct cylindrical regions with different flow regimes.

At intermediate times, the distance-drawdown curves in Figure 1 are subparallel to the early and late Theis curves within the range of β values between 0.001 and 7.0. Let us consider, for example, the particular dimensionless distance-drawdown curve corresponding to $\beta t_s = 10^2$. Since this curve is intermediate between the early and late Theis curves, it is clear that the cylindrical region of the aquifer confined between $r_D^2 = 0.001K_D^{-1}$ and $r_D^2 = 7.0K_D^{-1}$ is strongly influenced by vertical drainage (recall that $\beta = K_D r_D^2$). Thus if a region of predominantly artesian flow exists, as suggested by Gambolati, this region must occur at $r_D^2 \geq 7.0K_D^{-1}$. Furthermore, the existence of such a region would imply that our distance-drawdown curve must become asymptotic to the early Theis curve at sufficiently small values of dimensionless time. However, the steep portion of our curve is removed horizontally from the early Theis curve by approximately two logarithmic cycles and there is nothing in Figure 1 (nor in theory) to indicate that these two curves must necessarily meet. This implies that Gambolati's assumption about the perennial existence of a region within which the aquifer behaves in a predominantly artesian manner is not supported by evidence and thus remains subject to doubt.

A similar doubt is cast by theory about the validity of Gambolati's assumption that part of the aquifer will always respond according to the late Theis equation. As an example, let us consider the dimensionless distance-drawdown curve in Figure 1 which corresponds to $\beta t_s = 10^0$. This curve is far removed from the late Theis curve when β is between 0.001 and 7.0. Neither Figure 1 nor theory imply that these two curves must necessarily become asymptotic as dimensionless time increases, and the same would become even more evident if we would reduce the value of σ in Figure 1 below 10^{-4} (because then the late Theis curve would move to the right).

Thus there is nothing to indicate that at a time corresponding to $\beta t_s = 10^0$ there must always be a region in the aquifer which conforms to the late Theis equation; such a region may or may not exist, depending on the particular aquifer under study.

Figure 1 represents a practical range of β values that correspond to a cylindrical region between $r_D^2 = 0.001K_D^{-1}$ and $r_D^2 = 7.0K_D^{-1}$. The figure shows that when pumping starts, this entire region behaves as if it were artesian, but as time progresses, drawdowns everywhere in the region start deviating from early Theis behavior almost simultaneously. For example, at a relatively early time corresponding to $\beta t_s = 10^0$, the effect of vertical drainage is clearly starting to be felt at each point in the region. At a time corresponding to $\beta t_s = 10^1$, vertical drainage appears to dominate the response at each point. This situation persists for some time and is still felt when $\beta t_s = 10^2$. At this time, the effect of vertical flow starts slowly dissipating. The rate of dissipation increases with time and becomes significant everywhere in the region when $\beta t_s = 10^3$. After an additional amount of time, the effect of vertical drainage in the entire region dissipates almost completely as the time-drawdown curve asymptotically approaches the late Theis curve.

In summary, theory indicates that the effect of vertical drainage on the average drawdown is felt almost simultaneously everywhere in a large segment of the aquifer; there is no indication that the relative intensity of this effect varies significantly with radial distance from the pumping well. A similar conclusion has been reached by Neuman and Witherspoon [1972] about the relative effect of leakage on the drawdown response of a confined aquifer. Since the manifestation of vertical drainage in unconfined aquifers is very similar to that of leakage, the analogy between these two situations is not surprising.

PHYSICAL MEANING OF BOULTON'S α

Boulton's [1954b, 1963] theory is based on an integrodifferential equation of the general form

$$\nabla \cdot (T \nabla \bar{s}) = S \frac{\partial \bar{s}}{\partial t} + S_y \int_0^t \frac{\partial \bar{s}}{\partial t} \bigg|_{t=\tau} B(r, t - \tau) d\tau \quad (10)$$

where ∇ is the gradient operator in the horizontal plane. Boulton's assumption that the effect of specific yield decays exponentially with time leads to the following definition of the kernel:

$$B(t) = \alpha e^{-\alpha t} \quad (11)$$

where α is a property of the aquifer. By treating T , S , S_y , and α as constants, Boulton was able to solve (10) for the average drawdown around a completely penetrating well which is pumped at a constant rate Q . His solution can be expressed as

$$\bar{s}(r, t) = \frac{Q}{2\pi T} \int_0^\infty J_0(\mu_0 x) \left\{ 1 - e^{-\mu_1} \left[\cosh(\mu_2) + \frac{\mu_1}{\mu_2} \sinh(\mu_2) \right] \right\} \frac{dx}{x} \quad (12)$$

where

$$\mu_0 = r[(S + S_y)\alpha/T]^{1/2}$$

$$\mu_1 = \alpha t(1 + \sigma)^{-1}(1 - x^2)/2$$

$$\mu_2 = \alpha t[(1 + \sigma)^{-2}(1 + x^2)^2 - 4(1 + \sigma)^{-1}x^2]^{1/2}/2$$

In the particular case where $S = 0$, (12) reduces to the somewhat simpler form

$$\bar{s}(r, t) = \frac{Q}{2\pi T} \int_0^\infty J_0[\alpha(S_y\alpha/T)^{1/2}x] \left[1 - (1 + x^2)^{-1} \exp\left(-\frac{\alpha t x^2}{1 + x^2}\right) \right] \frac{dx}{x} \quad (13)$$

In (12) and (13), α is treated merely as an empirical constant to be determined from pumping test data. The possible relationship between α and other physical characteristics of the aquifer remains unclear in these equations. On the other hand, the definition of α in (10) and (11) does not necessarily require it to be a constant. However, since the integral in (10) has been derived intuitively rather than on the basis of physical considerations, the theory does not provide us with any clue as to the manner in which α might possibly vary with r or t .

The recent development of physically based models for delayed yield by Neuman [1972, 1973, 1974] and Streltsova [1972a, b, 1973] has led to the interesting question: to what extent is Boulton's empirical approach consistent with these new physical theories? In other words, if one assumes that the physical concepts underlying the theories of Neuman and Streltsova are correct, is it then possible to provide a consistent interpretation for Boulton's α in light of these new theories? Can these theories be used to assign a 'true' physical meaning to α ?

As was mentioned earlier, Boulton's analytical solutions (12) and (13) for the average drawdown can be reconciled with the equivalent solutions of Neuman's (5) and Streltsova merely by providing appropriate expressions for α in terms of other physical parameters together with the space coordinate r . For example, we mentioned that Streltsova's solution for the average drawdown becomes identical with Boulton's (12) and (13) when α is set equal to $3K_z/S_y b$ as in (3). On the other hand, to reconcile Boulton's (12) and (13) with Neuman's (5), one must express α in terms of K_z , Z_D , S_y , b , and r according to the solid curve in Figure 6 of Gambolati [1976]. For $0.01 \leq \beta \leq 5.00$, this curve can be closely approximated by (4). Since Neuman's solution is physically identical with that of Streltsova but is mathematically more accurate, we may conclude in accord with Neuman [1976] and Gambolati [1976] that in order to reconcile Boulton's (12) and (13) with the physically based solution one must allow α to vary with r . In other words, contrary to Boulton's assumption in deriving (12) and (13), α cannot be treated as a constant. We conclude that Boulton's treatment of α as a characteristic constant of the aquifer is inconsistent with the physically based approach; the fact that α

appears to be a constant in Streltsova's model is merely the result of a mathematical approximation. As a result of this mathematical inconsistency, the expressions for α in (4) and Gambolati's [1976] Figure 6 cannot be regarded as representations of the 'true' physical nature of α . Instead, these expressions must be viewed merely as empirical relationships which enable one to reconcile Boulton's (12) and (13) with the equivalent physically based solution of Neuman's (5).

Having recognized that (12) and (13) can only be used to obtain an empirical relationship between α and the physical characteristics of the aquifer, Gambolati [1976] has attempted to determine the 'true' physical meaning of α by approaching the problem in a different way. In his view [Gambolati, 1976, p. 32], 'the most natural way to investigate the actual physical meaning of α ' is to set Boulton's integrodifferential equation (10) for a constant T equal to the equation

$$T\nabla^2 \bar{s} = S \frac{\partial \bar{s}}{\partial t} + S_y \frac{\partial s}{\partial t} \Big|_{z=b} \quad (14)$$

This latter equation is obtained simply by integrating the governing equation of Neuman [1972, 1974]

$$K_r \left(\frac{\partial^2 s}{\partial r^2} + \frac{1}{r} \frac{\partial s}{\partial r} \right) + K_z \frac{\partial^2 s}{\partial z^2} = S_s \frac{\partial s}{\partial t} \quad (15)$$

with respect to z , subject to the linearized boundary condition at the water table [Neuman, 1972, 1974]

$$K_z \frac{\partial s}{\partial z} \Big|_{z=b} = -S_y \frac{\partial s}{\partial t} \Big|_{z=b} \quad (16)$$

When (10) with a constant T is set equal to (14), the result is

$$\int_0^t \alpha e^{-\alpha(t-\tau)} \frac{\partial \bar{s}}{\partial t} \Big|_{r=\tau} d\tau = \frac{\partial s}{\partial t} \Big|_{z=b} \quad (17)$$

Assuming that $\bar{s}$ and s in (17) can be computed from equations developed by Boulton [1954a] for a rigid water table aquifer which does not exhibit the elastic aspect of delayed yield (or from (5) by taking the limit as S tends to zero), Gambolati solved (17) for α as a function of K_r , K_z , S_y , b , r , and t . His results led him to conclude [Gambolati, 1976, p. 27] that

... the true physical α ... is almost linearly related to $1/r$ and becomes practically constant at some distance from the well (equal to twice the aquifer thickness if the medium is isotropic).

Referring to Neuman's [1975a] attempt at reconciling Boulton's analytical solution with his own through the empirical relationship (4), Gambolati [1976, p. 35] criticized Neuman for

... [failing] to provide a constructive response to this important problem, since ... he seems to confuse conceptually the empirical $[\alpha]$ with the actual physical $[\alpha]$ and a reader is inevitably led to identify the former and its behavior with the latter.

The difficulty with Gambolati's viewpoint is that it fails to recognize the inherent mathematical inconsistency between Boulton's operational approach and the physically based theory of Neuman. The mathematical inconsistency between Boulton's analytical solutions (12)–(13) and Neuman's (5) has already been demonstrated in the preceding paragraphs. To show that the same holds true with regard to Boulton's integrodifferential equation (10) and Neuman's governing equations (15)–(16), one must investigate the conditions under which (15)–(16) can be transformed into such an integrodifferential form. Since (15) and (16) lead to (14), it is sufficient to investigate the conditions under which (10) can be set equal to (14), which forms the basis for Gambolati's conclusions.

An investigation of these conditions has been performed recently by *Herrera et al.* [1978]. These authors were able to show very clearly [*Herrera et al.*, p. 212] that a relation of the form

$$\int_0^t B(r, t - \tau) \frac{\partial s}{\partial t} \bigg|_{t=\tau} d\tau = \frac{\partial s}{\partial t} \bigg|_{z=b} \quad (18)$$

'can hold only if'

$$K_r b \left(\frac{\partial^2 s^*}{\partial r^2} + \frac{1}{r} \frac{\partial s^*}{\partial r} \right) \ll K_z \left(b \frac{\partial^2 s^*}{\partial z^2} - \frac{\partial s^*}{\partial z} \bigg|_{z=b} \right) \quad (19)$$

where $s^* = s - \bar{s}$. However, as I shall demonstrate in the following section, (19) is generally not satisfied in the vicinity of the pumping well and is satisfied far from the well only at relatively large values of time. This implies that Boulton's integrodifferential equation (10) is mathematically inconsistent with the physically based equations (15)–(16) of Neuman, and therefore one cannot assign any 'true' physical meaning to α . The best that one can do is compensate for this inconsistency by deriving another empirical expression for α in a manner analogous to that of *Gambolati* [1976] to supplement the earlier empirical expression derived by *Neuman* [1975]. Any argument as to which of these two expressions is theoretically more correct than the other is futile, because both are the results of attempts to reconcile two different theories that are mathematically inconsistent.

VALIDITY OF INTEGRODIFFERENTIAL EQUATION FOR DELAYED YIELD

Herrera et al. [1978] have shown that the average drawdown in an unconfined aquifer can be represented by an integrodifferential equation such as (10) provided that (19) is satisfied. Thus it is important to investigate the conditions under which (19) holds true. A partial answer to this question can be immediately provided for the special case where $S = 0$. In this case, the theory of *Herrera et al.* [1978, equations (24)–(25)] implies that if (19) is correct, then the kernel in (10) assumes the form $B(t) = \alpha e^{-\alpha t}$ as in (11) with $\alpha = 3K_z/S_y b$ as in (3). On the other hand, *Gambolati* [1976] has shown that when $S = 0$ and $B(t) = \alpha e^{-\alpha t}$, the integrodifferential approach cannot be reconciled with the physically based theory of Neuman unless

$$\alpha = \epsilon K_z/S_y b \quad (20)$$

where ϵ is a function of β and dimensionless time [*Gambolati*, 1976, Table 1]. For example, when $t_y = 10^{-4}$, ϵ can be approximated by

$$\epsilon = 2.4 + \{0.384\beta^{-0.448}\} \quad (21)$$

Since β is proportional to r^2 , (21) indicates that α varies almost linearly with r in the vicinity of the pumping well, and it becomes equal to $2.4K_z/S_y b$ at a sufficient distance away from the well. However, for condition (19) to be true, α must remain constant and equal to $3K_z/S_y b$ (i.e., ϵ must be equal to 3). Table 1 of *Gambolati* [1976] clearly shows that this is not the case: ϵ is much larger than 3 near the pumping well, and it does not attain a constant value of 3 farther away from the well unless dimensionless time is sufficiently large. For example, (21) shows that when $t_y = 10^{-4}$, ϵ tends to a constant value of 2.4 as radial distance from the well increases, and it is equal to 3 only at one particular value of r . Thus it is clear that the condition required for the integrodifferential approach to be

mathematically consistent with Neuman's physically based theory is not satisfied unless r and t are sufficiently large.

EFFECTIVE STORAGE COEFFICIENT AT LARGE VALUES OF TIME

Neuman's [1972, 1973] solution for point drawdown in the case of a fully penetrating pumping well is given by

$$s(r, z, t) = \frac{Q}{4\pi T} \int_0^\infty 4x J_0(x\beta^{1/2}) \left[w_0(x) + \sum_{n=1}^\infty w_n(x) \right] dx \quad (22)$$

where

$$w_0(x) = \frac{\{1 - \exp[-t_y\beta(x^2 - \gamma_0^2)]\} \cosh(\gamma_0 z_D)}{\{x^2 - (1 + \sigma)\gamma_0^2 - [(x^2 - \gamma_0^2)^2/\sigma]\} \cosh(\gamma_0)} \quad (23)$$

$$w_n(x) = \frac{\{1 - \exp[-t_y\beta(x^2 + \gamma_n^2)]\} \cos(\gamma_n z_D)}{\{x^2 - (1 + \sigma)\gamma_n^2 - [(x^2 + \gamma_n^2)^2/\sigma]\} \cos(\gamma_n)} \quad (24)$$

and the quantities γ_0 and γ_n are the roots of (8) and (9). *Neuman* showed that at early time (22) reduces to the following form of the Theis equation:

$$s(r, t) = \frac{Q}{4\pi T} \int_{1/4t_y}^\infty \frac{e^{-y}}{y} dy \quad (25)$$

This is equivalent to saying that the aquifer behaves as if it were fully confined with a storage coefficient S . At large values of time, his former analysis suggested that (22) should reduce to

$$s(r, t) = \frac{Q}{4\pi T} \int_{1/4t_y}^\infty \frac{e^{-y}}{y} dy \quad (26)$$

which is another form of the Theis equation. This latter form implies that the effect of elastic storage essentially dissipates with time. In other words, (26) means that at large t the water is released from storage only by gravity drainage, and the final storage coefficient is S_y .

Gambolati [1976] pointed out rightly that *Neuman's* derivation of (26) from (22) is rigorously correct only if S is negligible compared to S_y . For the more general case where this is not so, *Gambolati* proved that the average drawdown at large values of time is given by

$$\bar{s}(r, t) = \frac{Q}{4\pi T} \int_{1/4t_y}^\infty \frac{e^{-y}}{y} dy \quad (27)$$

This form of the Theis equation implies that the effect of elastic storage does not dissipate with time. Instead, the effects of elastic storage and gravity drainage become additive at large t , the final storage coefficient being simply the sum of S and S_y . *Gambolati* [1976] also proved that the drawdown at the water table approaches $\bar{s}$ as time increases.

It is easy to show that (27) is consistent with *Neuman's* theory and can in fact be generalized to the case of point drawdowns anywhere in the aquifer. To show this, note that for large t the value of the integral in (22) is determined primarily by the small values of x . Since according to (8) and (9) one always has $\gamma_0^2 < x^2$ and $\gamma_n > \pi/2$ for $n \geq 1$, it is seen that for small x and large t , (6) and (7) tend asymptotically to

$$w_0(x) \sim \frac{1 - \exp[-t_y\beta(x^2 - \gamma_0^2)]}{x^2 + (1 + \sigma)\gamma_0^2} \quad (28)$$

$$w_n(x) \sim 0 \quad n \geq 1 \quad (29)$$

Similarly, the asymptotic form of (8) is

$$\sigma\gamma_0^2 - (x^2 - \gamma_0^2) \sim 0 \quad (30)$$

because when γ_0 is small, $\sinh(\gamma_0) \sim \gamma_0$ and $\cosh(\gamma_0) \sim 1$. Solving (30) for γ_0^2 , substituting into (28), and then combining (28) and (29) with (22), one obtains

$$s(r, t) = \frac{Q}{2\pi T} \int_0^\infty J_0(x\beta^{1/2}) [1 - \exp(-t_{sy}\beta x^2)] \frac{dx}{x} \quad (31)$$

From tables of definite integrals [Gradshteyn and Ryzhik, 1963, p. 731], one can find

$$\int_0^\infty \exp(-ax^2) J_0(x) \frac{dx}{x} = \frac{1}{2} \int_0^{1/4a} \frac{e^{-x}}{x} dx \quad (32)$$

where the real part of a is nonnegative. When (32) is combined with (31), one immediately obtains

$$s(r, t) = \frac{Q}{4\pi T} \int_{1/4t_{sy}}^\infty \frac{e^{-y}}{y} dy \quad (33)$$

which shows that the drawdown no longer varies with the vertical coordinate z . Thus (33) implies that according to Neuman's theory, the average drawdown is given by (27), as suggested earlier by Gambolati [1976].

The above result is of practical interest only in the case of relatively thick aquifers in which the magnitude of S approaches or exceeds that of S_y . In such situations, the time-drawdown response exhibits only a very slight inflection [Neuman, 1972; Gambolati, 1976, Figure 11], and the aquifer behaves almost as if it were confined. In situations in which the effect of delayed yield is clearly evident, S is much smaller than S_y , and therefore (11) and (12) become practically identical.

CONCLUSIONS

The foregoing discussion leads to the following conclusions:

1. Boulton's delayed yield theory, in which the specific yield varies exponentially with time, does not in any direct way account for the release of water from storage in the unsaturated zone. Instead, this 'operational' theory can be made to fit other physically based models which account for the release of water from artesian storage beneath the water table together with gravity drainage of the saturated zone as the water table drops.

2. Air entry effects of the kind described recently by Bouwer and Rice [1978] may, perhaps, play a role in the delayed yield process. However, the importance of this role in comparison to that of artesian storage does not appear to be entirely clear at the present time. The air entry concept leads to theoretical difficulties which have not yet been completely resolved and will require further investigation. Furthermore, there is neither theoretical nor experimental evidence to support the view that air entry effects are capable of producing an inflection in the time-drawdown curve of unconfined aquifers when compressibility is eliminated from the model.

3. In analyzing time-drawdown data from wells which fully penetrate an unconfined aquifer, the models of Boulton [1954b, 1963], Neuman [1972, 1973, 1974, 1975a], and Streltsova [1972a, b, 1973] will yield practically identical values of T , S , and S_y (or K_r , S_a , and S_y , if b is known). The model of Neuman will also yield a value for K_z which can be expected to be more accurate than that obtained from Streltsova's model. Boulton's model, on the other hand, will not yield a value for K_z but only the lumped parameter α . All three models are applicable to the same class of unconfined flow problems.

4. In analyzing time-drawdown data from partially penetrating wells, the models of Boulton [1954b, 1963] and Streltsova [1972a, b, 1973] are no longer applicable, and one must use the method of Neuman [1974, 1975a].

5. The effect of vertical drainage on the average drawdown is felt almost simultaneously everywhere in a large portion of the aquifer; there appears to be little indication that the relative intensity of this effect must always vary sufficiently with radial distance from the pumping well in order to justify dividing the aquifer into three cylindrical regions such that only the middle region is influenced by vertical flow, as suggested by Gambolati [1976].

6. Boulton's [1954b, 1963] delayed yield theory is mathematically inconsistent with the linearized theory of Neuman [1972, 1973, 1974]. For this reason, Boulton's α cannot be assigned any 'true' physical meaning in light of Neuman's theory. The best that one can do is obtain empirical expressions for α in terms of other physical parameters together with the space and time coordinates, as has been done in different ways by Neuman [1975a] and Gambolati [1976]. Each of these empirical expressions is valid in its own right, and none can be claimed to represent a unique scientific truth.

7. The conditions required in order to represent unconfined flow by consistent integrodifferential equations are generally not satisfied in the vicinity of a pumping well and are satisfied away from this well only at sufficiently large values of time.

8. The effect of elastic storage in an unconfined aquifer does not dissipate with time. Instead, the effects of elastic storage and gravity drainage become additive at large t , the final storage coefficient being the sum of S and S_y . This result, originally pointed out by Gambolati [1976], is of practical interest only in the case of relatively thick aquifers in which the magnitude of S approaches or exceeds that of S_y . However, in such situations the aquifer behaves almost as if it were confined. In situations where the effect of delayed yield is clearly evident, S is much smaller than S_y , and therefore the influence of S at large values of time can be safely neglected.

NOTATION

b	initial saturated thickness of aquifer, L .
$J_0(x)$	zero-order Bessel function of the first kind.
K_D	dimensionless permeability, equal to K_z/K_r .
K_r	horizontal permeability, LT^{-1} .
K_z	vertical permeability, LT^{-1} .
Q	pumping rate, L^3T^{-1} .
r	radial distance from pumping well, L .
S	storage coefficient, equal to $S_a b$.
s	drawdown, L .
$\bar{s}$	drawdown averaged over the vertical, L .
s_D	dimensionless drawdown, equal to $4\pi Ts/Q$.
$\bar{s}_D$	dimensionless drawdown averaged over the vertical.
S_a	specific (elastic) storage, L^{-1} .
s_{wt}	drawdown of the water table, L .
S_y	specific yield.
T	transmissivity, $K_r b$, L^2T^{-1} .
t	pumping time, t .
t_s	dimensionless time with respect to S_a , equal to Tt/Sr^2 .
t_{sy}	dimensionless time with respect to S_a and S_y , equal to $Tt/(S + S_y)r^2$.
t_y	dimensionless time with respect to S_y , equal to $Tt/S_y r^2$.
z	vertical distance above bottom of aquifer, L .
z_D	dimensionless elevation, equal to z/b .

- α reciprocal of Boulton's delay index, T^{-1} .
 β $K_D r^2 / b^2$.
 σ S/S_y .

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